Scenario 1: Intermittent communication loss from a satellite

Problem: A satellite sometimes loses contact with its ground station, making it difficult for the team to receive telemetry and send commands.

1. Descriptive — What happened?

Communication logs show that the satellite lost contact several times during the past week. Most interruptions lasted a few minutes, but some lasted longer than expected.

2. Diagnostic — Why did it happen?

Comparing communication logs with satellite position, signal strength, and ground-station records may show that the losses happen more often at certain orbital positions or when signal quality is poor. The team should check these possibilities before deciding on the cause.

3. Predictive — What might happen next?

If the same pattern continues, the satellite may experience more communication interruptions during similar parts of its orbit. Longer interruptions could delay the delivery of mission data.

4. Prescriptive — What should we do now?

Review the times and locations of previous interruptions, check antenna and ground-station performance, and adjust communication schedules if needed. The team should also make sure important telemetry is stored safely until contact is restored.

Scenario 2: Increased risk from orbital debris

Problem: A spacecraft is operating in an area where several pieces of space debris have been identified, creating a possible collision risk.

1. Descriptive — What happened?

Tracking data shows that the number of debris objects passing near the spacecraft's orbit has increased. Some objects have estimated paths that bring them relatively close to the spacecraft.

2. Diagnostic — Why did it happen?

The team compares the predicted paths, object sizes, tracking accuracy, and orbital data to understand which objects may pose a real risk. Some close approaches may be harmless, while others need closer monitoring.

3. Predictive — What might happen next?

If the predicted paths remain similar, some debris objects may continue to pass close to the spacecraft. New tracking information could change the estimated risk, so the predictions need regular updates.

4. Prescriptive — What should we do now?

Continue monitoring the objects, update collision-risk calculations, and follow the mission's approved procedures if a close approach requires action. Any manoeuvre should be reviewed carefully because it can affect fuel use and the spacecraft's planned orbit.

Scenario 3: A Mars rover's energy level is decreasing

Problem: A Mars rover is using more energy than expected, leaving less power available for driving, communication, and scientific experiments.

1. Descriptive — What happened?

Power records show that the rover's available energy has decreased over several days. Some planned activities have used more power than expected, and the rover has less energy remaining at the end of each day.

2. Diagnostic — Why did it happen?

The team compares power consumption with driving distance, instrument use, temperature, and battery performance. This can help identify whether the higher usage is related to environmental conditions, equipment, or the daily activity plan.

3. Predictive — What might happen next?

If energy use continues at the same rate, the rover may have to reduce some activities to protect its power supply. It could also have less energy available for communication or experiments on days with difficult conditions.

4. Prescriptive — What should we do now?

Plan activities according to the available power, prioritize the most important scientific tasks, and reduce unnecessary energy use. The team should keep checking battery and power-system readings and follow the mission's operating limits.

Scenario 4: Astronaut fatigue during a long-duration mission

Problem: Astronauts on a long-duration space mission may experience fatigue that affects their concentration, sleep, and ability to complete tasks.

1. Descriptive — What happened?

Crew schedules and sleep records show that some astronauts have been sleeping less than planned and reporting higher levels of tiredness. The pattern appears more often during busy periods with several tasks scheduled close together.

2. Diagnostic — Why did it happen?

The team reviews sleep schedules, workload, exercise, and reports from the crew to look for possible reasons. A crowded schedule or interruptions to sleep may contribute, but the records alone may not prove a single cause.

3. Predictive — What might happen next?

If the workload and sleep pattern continue, fatigue may increase and make demanding tasks harder to complete. This is a possibility rather than a certainty, so the crew's condition should continue to be monitored.

4. Prescriptive — What should we do now?

Review the work schedule, protect planned sleep periods, and spread demanding tasks where possible. Crew health and safety procedures should guide any changes, and astronauts should be encouraged to report fatigue early.

Scenario 5: A deep-space probe has a growing data backlog

Problem: A deep-space probe collects more scientific data than it can send to Earth during its available communication windows.

1. Descriptive — What happened?

Mission records show that the probe's stored data has increased over time. During some communication windows, the amount of data transmitted to Earth is lower than the amount collected by the instruments.

2. Diagnostic — Why did it happen?

The team compares the volume of collected data with communication-window duration, signal conditions, transmission rates, and ground-station availability. This helps determine whether the backlog is mainly caused by limited contact time, low transmission speed, or unusually high data collection.

3. Predictive — What might happen next?

If data continues to arrive faster than it can be transmitted, the backlog may keep growing. If the storage capacity becomes limited, the probe may eventually need to change how it collects or stores new observations.

4. Prescriptive — What should we do now?

Prioritize the most valuable scientific data, schedule communication opportunities carefully, and use suitable data compression where possible. The team should monitor storage capacity and adjust observation plans when necessary, following mission requirements.